PULSEIQ — PHASE 3 SUBMISSION REPORT

Workforce Analytics Application | Data Analytics | 6 November 2026

Prepared by	Dilip D	**GitHub**	github.com/Dilip111004/pulseiq
Phase tag	phase-3	**Branch**	phase-3-money-and-insight
Release	Phase 3 tag pushed to GitHub; Phase 3 README configured for tag view.	**Live app**	Open PulseIQ Live Dashboard

Executive summary

PulseIQ is a Streamlit-based workforce analytics application that turns workforce, payroll and operational data into management dashboards. Phase 3, titled **Money and Insight**, adds financial performance and management insight through P&L and expense analytics, invoices and receivables, cash-flow indicators, cross-department impact, AI signals and an executive scorecard. The Phase 3 application is deployed publicly through Streamlit Community Cloud.

The Phase 3 implementation is maintained on the dedicated **phase-3-money-and-insight** branch. The **phase-3** tag was pushed to GitHub. The Phase 1 and Phase 2 work was preserved on their respective branches/tags.

1. Phase 3 task delivery

TASK	DASHBOARD / MODULE	IMPLEMENTATION EVIDENCE
Day 11	P&L and Expense Analytics	Page 13 implemented and validated. Operating revenue ■125.92 Cr; total expenses ■103.63 Cr; net profit ■22.47 Cr; operating margin 17.70%.
Day 12	Invoices, Receivables and Cash Flow	Page 14 implemented and validated. 62 open invoices; open receivables ■13.94 Cr; payment and ageing summaries checked.
Day 13	Cross-Department Impact	Page 9 completed. 198 department-month rows; 11 departments; all 198 keys matched; no duplicate keys or missing department names/employer costs.
Day 14	AI Signals	Page 10 completed. 405 anomaly rows flagged; money at risk ■3.14 Cr; forecast covers Oct 2026–Mar 2027.
Day 15	Executive Scorecard	Page 11 completed. Eight selected KPIs with month selector, targets, trends/status and links to detailed analysis.

2. Additional analytics and reporting

MODULE	RECORDED RESULT
P&L and financial metrics	Operating revenue ■1,259,204,600; other income ■1,725,452; total expenses ■1,036,264,914.42; operating profit ■222,939,685.58; net profit ■224,665,137.58; net profit margin 17.82%.
Receivables and payments	Open receivables ■139,394,278; 708 payments recorded; payments received ■1,418,299,830. Paid-expense cash outflow ■185,647,035.90 (paid expense records only).
AI signals and forecast	405 anomaly records; ■31,421,239.07 money at risk across 178 records with positive amounts. Forecast horizon Oct 2026–Mar 2027.

LIVE STREAMLIT DASHBOARD
Open the deployed PulseIQ Phase 3 application:
Open PulseIQ Live Dashboard

3. Validation and operational evidence

AREA	RECORDED RESULT
Cross-department reconciliation	198 department-month records reconciled across Apr 2025–Sep 2026 and 11 departments; 198/198 keys matched; no duplicate keys. Attendance-rate max difference 0.00005449 (tolerance 0.0001); cost-per-head max difference 0.005 (tolerance 0.01).
P&L and receivables	Financial totals and receivables ageing summaries were validated during Phase 3 checks. Open receivables ■13.94 Cr; overdue ageing buckets: 1–30 days ■22,133,626; 31–60 days ■0; 61–90 days ■1,940,982; 90+ days ■3,036,258.
AI Signals	405 anomaly rows generated and flagged; money at risk ■31,421,239.07. These are investigation signals, not confirmed losses or proof of fraud.
Forecast evaluation	Forecast trained through Sep 2026 for Oct 2026–Mar 2027. Actual observations for the forecast horizon were not available, so forecast accuracy / MAPE could not be measured.
Executive scorecard	Eight selected KPIs populated with target/trend/status presentation and navigation to detailed pages. Financial thresholds are management-proposed comparison targets, not formally approved corporate targets.
Cloud deployment	DuckDB startup initialization fix deployed. User confirmed PQ-11 and PQ-12 were working correctly after redeployment.
GitHub delivery	Phase 3 branch **phase-3-money-and-insight** and **phase-3** tag pushed. Phase 3 README made available as the README shown when opening the phase-3 tag.

4. Problems encountered and resolutions

Cloud database initialization: The deployed PQ-11 and PQ-12 pages initially could not find the DuckDB database. The Phase 3 app was updated to test database usability and rebuild the generated warehouse from tracked data when the database was missing or unusable. After redeployment, the pages worked.

Forecast accuracy unavailable: The forecast period had no overlapping actual values. The limitation was documented and no unsupported MAPE or accuracy figure was claimed.

Metric reconciliation differences: Department-month output was compared with the reference using join keys and metric tolerances. All 198 keys matched, and the observed attendance-rate and cost-per-head differences were within the stated tolerances.

Financial target provenance: Financial KPI thresholds were identified as management-proposed targets rather than approved corporate targets, and are described that way in the report.

5. Key learning

• I learned to validate financial totals and receivables at the correct grain before presenting them as management KPIs.
• I strengthened my ability to reconcile department-month data using keys and explicit metric tolerances.
• I learned to explain forecast limitations honestly when actual values are not yet available.
• I gained practical experience debugging cloud database initialization and maintaining Git branches, tags and release documentation.

6. Phase review and next-phase direction

Mini challenge — management question: “What is the financial position, and where should management investigate further?” A short walkthrough can use Page 13 for revenue, expenses and profit; Page 14 for open receivables and ageing; and Page 10 for anomaly signals and money at risk. Page 9 adds department-level employer and overtime cost context. A specific cause for a monthly cost increase should only be stated after a month-specific drill-down confirms the driver.

Self-check: The department-month validation demonstrates matching keys and metric comparisons. In a review, I should be able to explain each source's grain, join keys and how the displayed total is reconciled.

Next phase: Evaluate forecasts when actuals become available; replace proposed thresholds with approved targets; strengthen month-level cost-driver analysis; retain screenshots and checker outputs for evidence; and continue cloud deployment and regression checks.

Repository: github.com/Dilip111004/pulseiq
Branch: phase-3-money-and-insight
Tag: phase-3
Live dashboard: Open PulseIQ Live Dashboard